# Mahmoud Alyosify

Specialised Reinforcement Learning Engineer | Deep RL for Sequential Decision-Making & Resource Optimisation | MSc Artificial Intelligence, Queen's University (Canada)

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## PROFESSIONAL SUMMARY

Reinforcement Learning Engineer building agents that make **sequential decisions under uncertainty with real cost consequences**. Designed a **custom Gymnasium environment** modelling cloud workload dynamics and trained **PPO** and **DQN** agents to optimise a **cost–latency reward**, benchmarked honestly against static and threshold-based baselines rather than against a weak strawman. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, with graduate coursework in **Reinforcement Learning, Deep Learning and Cloud Computing**. Complementary experience in **multi-agent orchestration** and **decoding-time policies for language models**, plus national-scale systems engineering. Seeking RL roles in control, resource optimisation, simulation and agentic decision systems.

## EDUCATION

**MSc in Artificial Intelligence** — *Sep 2025 – Nov 2026*
*Queen's University, Kingston, Ontario, Canada* — Expected Nov 2026

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Coursework: **Reinforcement Learning, Deep Learning, Cloud Computing, Large Language Models, End-to-End ML Pipelines**.
- **Master's Research Project (CISC 898):** decoding-time control policies for token-efficient language-model inference — Supervisor: Dr. Rahatara Ferdousi.

**B.Sc. Computer & Information Science (Bioinformatics)** — *Sep 2019 – Jul 2023*
*Assiut University, Egypt* — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

## TECHNICAL SKILLS

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|---|---|
| **Reinforcement Learning** | Deep Reinforcement Learning, PPO, DQN, Policy-Gradient & Value-Based Methods, Custom Gymnasium Environment Design, Reward Shaping, Exploration–Exploitation Trade-offs, Multi-Objective Reward Design, Baseline Benchmarking, Policy Evaluation |
| **Simulation & Systems** | Workload Simulation, Cloud Resource Provisioning, Cost–Latency Optimisation, Autoscaling Policies, Multi-Agent Systems, Agentic Workflows |
| **Deep Learning** | PyTorch, TensorFlow, CNNs, Transformers, Self-Supervised & Contrastive Learning, Fine-Tuning (LoRA, QLoRA), Optuna Hyperparameter Search |
| **Engineering & MLOps** | Python, NumPy, Pandas, AWS, Docker, FastAPI, ONNX, CI/CD, Git/GitHub, Linux, Reproducible Experiment Design, Latency & Energy Profiling |
| **Programming** | Python, C++, C, SQL, C#, Go, MATLAB, Bash |

## SELECTED REINFORCEMENT LEARNING PROJECTS

**RL-Cloud-Autoscaler — Autonomous Cloud Resource Provisioning via Deep RL** — *2026*
*Python · PPO · DQN · Gymnasium · Cloud Cost Optimisation*

- Designed and implemented a **custom Gymnasium environment** simulating cloud workload dynamics — defining the state space, discrete provisioning action space and episode structure from scratch.
- Engineered a **multi-objective reward** capturing the **cost–latency trade-off**, so the agent is penalised for both over-provisioning spend and SLA-breaching latency.
- Trained and tuned **PPO** (policy-gradient) and **DQN** (value-based) agents, comparing sample efficiency, stability and final policy behaviour under fluctuating demand.
- Benchmarked both agents against **static allocation** and **reactive threshold-based autoscaling** baselines — the policies real infrastructure teams actually run — to quantify genuine improvement.

**Verbosity-Aware Decoding — Learned Stopping Policy for LLM Inference** — *2026 – Present*
*MSc Research Project (CISC 898), Queen's University · Sequential Decision-Making · PyTorch*

- Formulated generation as a **step-by-step control problem**: at every decoding step an **entropy-driven gate** decides whether to continue or terminate, fusing stop-token, n-gram-repetition and semantic-redundancy signals — training-free, with no architecture change.
- Evaluated the resulting policy across **three model families** (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) on **MT-Bench, XSum** and **SQuAD-v2**, measuring the reward-relevant trade-off directly: token savings and latency against a non-inferiority bound on output quality.

**HORUS Sentinel — Autonomous Multi-Agent Intelligence Platform** — *2026*
*Multi-Agent Systems · Autonomous Agents · Knowledge Graphs · Self-Hosted LLM*

- Architected a **swarm of specialised autonomous agents** that continuously act on an environment of open data sources, coordinating through a shared knowledge graph rather than a central controller.

### Additional Projects

- **SCRAP** — risk-prediction pipeline supporting time-critical decisions, forecasting satellite collision risk **48+ hours** before closest approach.
- **SimCLR-Vision-SSL** — self-supervised representation learning with a **42-experiment** sweep, **84.30% top-1**, ONNX/FAISS deployment.
- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** records and deployed it on AWS.
- **Also:** Automated Multimodal Agent · AudioShield · Vitalism Solution (rPPG).

## PROFESSIONAL EXPERIENCE

**Machine Learning Instructor** — *Jul 2025 – Sep 2025*
*National Telecommunication Institute (NTI)* — Cairo, Egypt

- Delivered a **90-hour** hands-on ML program covering supervised learning, PCA and neural networks, with practical lab sessions throughout.

**AI in Cybersecurity Intern** — *Jun 2025 – Jul 2025*
*e& Egypt — via NTI (On-the-Job Training)* — Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

**Systems Engineer (Military Service)** — *Feb 2024 – Mar 2025*
*National Company for Roads Building & Development* — Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**; debugged and optimised real-time transportation software.

**Online Instructor — AI, Algorithms & Data Science** — *2020 – Present*
*Udemy & YouTube* — Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

## CERTIFICATIONS & AWARDS

**Machine Learning Specialization** — DeepLearning.AI
**HCIP – AI** (Huawei)
**Digilians Presidential Scholarship** — Queen's University
**Semi-Finalist** — Microsoft Imagine Cup 2023
**AWS Academy** — ML Foundations, ML for NLP, Data Engineering
**ML Internship** — ITIDA / Egypt Makes Electronics (200 hrs)
**1st Place** — Smart Cities Hackathon 2022
**Ideal Student Award** — Assiut University, 2022–2023

## LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.